AI-Based Anti-Money Laundering (AML) Systems in Digital Banking: A Hybrid Transactional and Graph-Based Approach

PROFESSOR

Ahmed El Seddawy

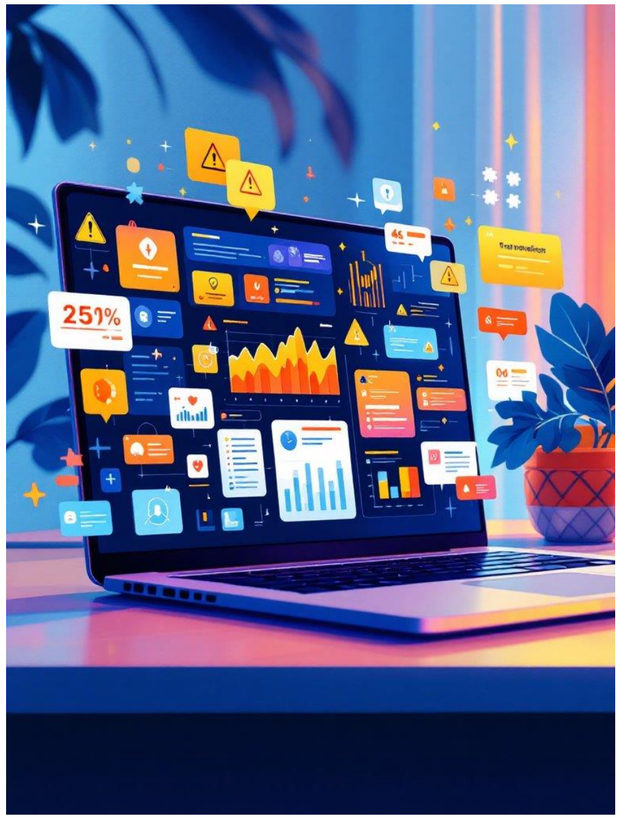
PRESENTED BY

Omnia Mounir (23226236)

Industry Context

The rapid growth of digital banking has created significant **money laundering challenges**. With complex schemes and increasing regulatory pressure, innovative AML solutions are urgently needed.

Digital payments, mobile banking, and crypto have increased transaction volume and complexity. FATF estimates money laundering at 2–5% of global GDP. Regulators demand real-time monitoring and auditability (GDPR, AMLD); legacy rule-based AML systems struggle to scale and adapt to networked laundering techniques



Problem Statement

Rule-based engines produce high alert volumes and up to 90–95% false positives, failing to detect layering, structuring, and network obfuscation. This creates operational overload, high investigation costs, and regulatory risk. The need: adaptive models that analyze both transaction attributes and entity relationships in near real-time.

Traditional AML Systems

FALSE POSITIVES

Traditional AML systems generate up to **95% false positives**, leading to unnecessary investigations and wasted resources, which ultimately undermines the efficiency of the overall detection process.

COMPLEX LAUNDERING SCHEMES

Money laundering schemes often involve **layering** and **network transactions**, making them difficult to detect with conventional methods, allowing illicit activities to thrive unnoticed in the financial system.

REGULATORY PRESSURE

As regulatory scrutiny increases, financial institutions face demands for **sophisticated AML solutions** that can adapt to evolving laundering tactics and ensure compliance with stringent legal requirements.

Limitations of Traditional AML

Traditional AML systems struggle to adapt to evolving laundering patterns. Their rule-based approach leads to excessive false positives and high operational costs, necessitating AI-driven solutions for better accuracy.

Data Sources Overview

The datasets utilized in developing the AI-based AML system, highlighting their significance in providing comprehensive insights into transaction behaviors and network relationships.

PAYSIM DATASET

The PaySim dataset includes various **transaction types**, amounts, balances, and fraud labels, providing key insights into mobile money transactions and their behavioral patterns for analysis.

ELLIPTIC DATASET

The Elliptic dataset consists of a **graph structure** displaying nodes representing transactions and edges indicating relationships, facilitating the clustering of transaction patterns to detect money laundering activities.

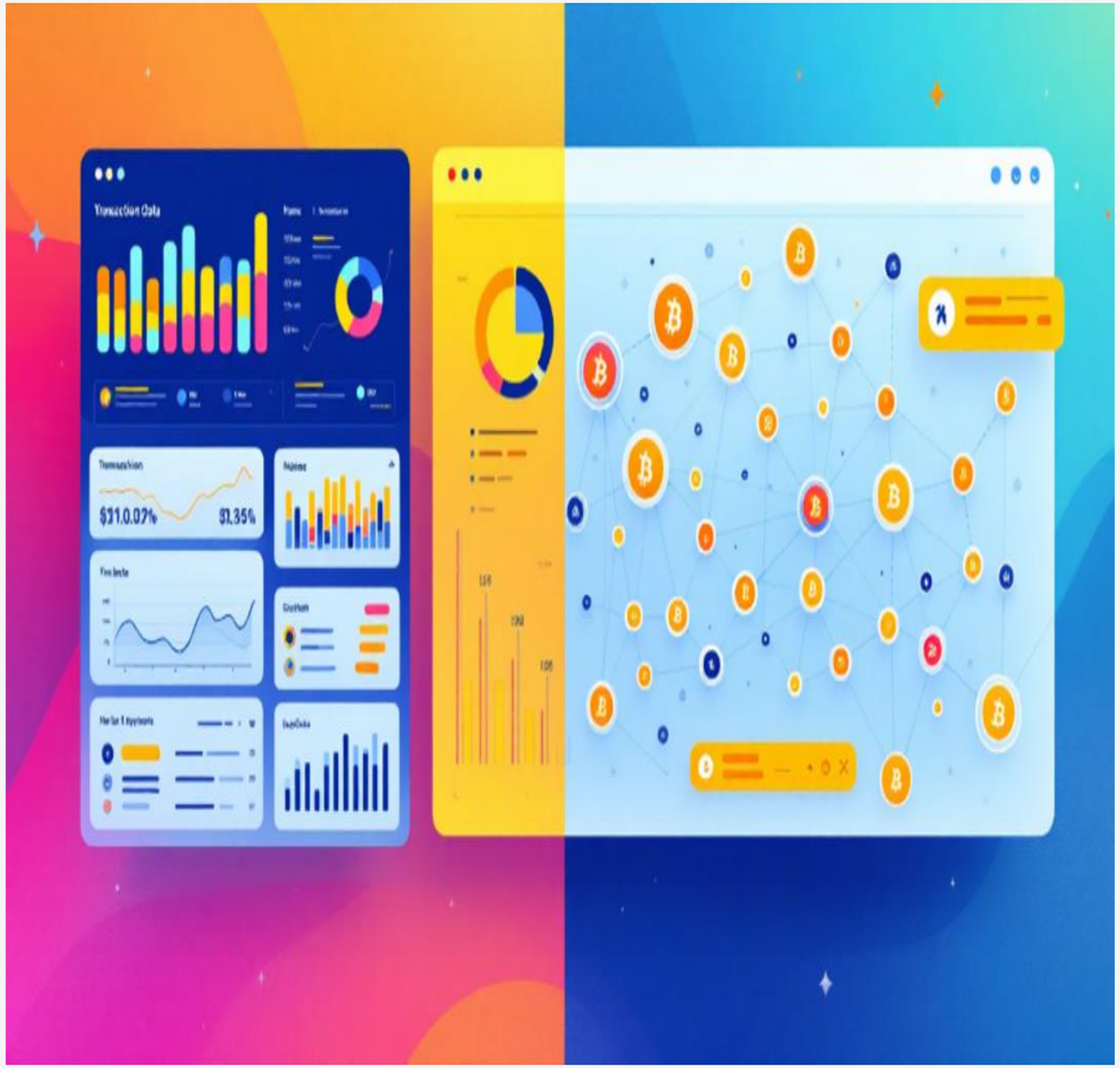
Data Strategy — Dual Datasets

The datasets utilized in developing the AI-based AML system, highlighting their significance in providing comprehensive insights into transaction behaviors and network relationships.

Hybrid dataset approach aligns network and behavioral analysis:

PaySim:~6M simulated transactions for behavioral/time-series modeling.

Elliptic:~200k Bitcoin transactions and graph structure for network detection. Both datasets include structured, temporal, and graph features enabling multi-layer analytics.



AI Techniques Overview

SUPERVISED LEARNING

Supervised machine learning classifies transactions based on labeled data, enhancing detection of suspicious activities by learning patterns from historical transactions to improve accuracy and effectiveness.

ANOMALY DETECTION

Unsupervised anomaly detection identifies unknown patterns by analyzing transaction data without predefined labels, enabling the system to discover novel laundering strategies that may otherwise go unnoticed.

GRAPH NEURAL NETWORKS

Graph Neural Networks (GNNs) analyze complex relationships between entities, revealing hidden laundering networks. They enhance the system's capacity to detect intricate schemes that traditional methods often overlook.

AI Architecture Overview

Key components

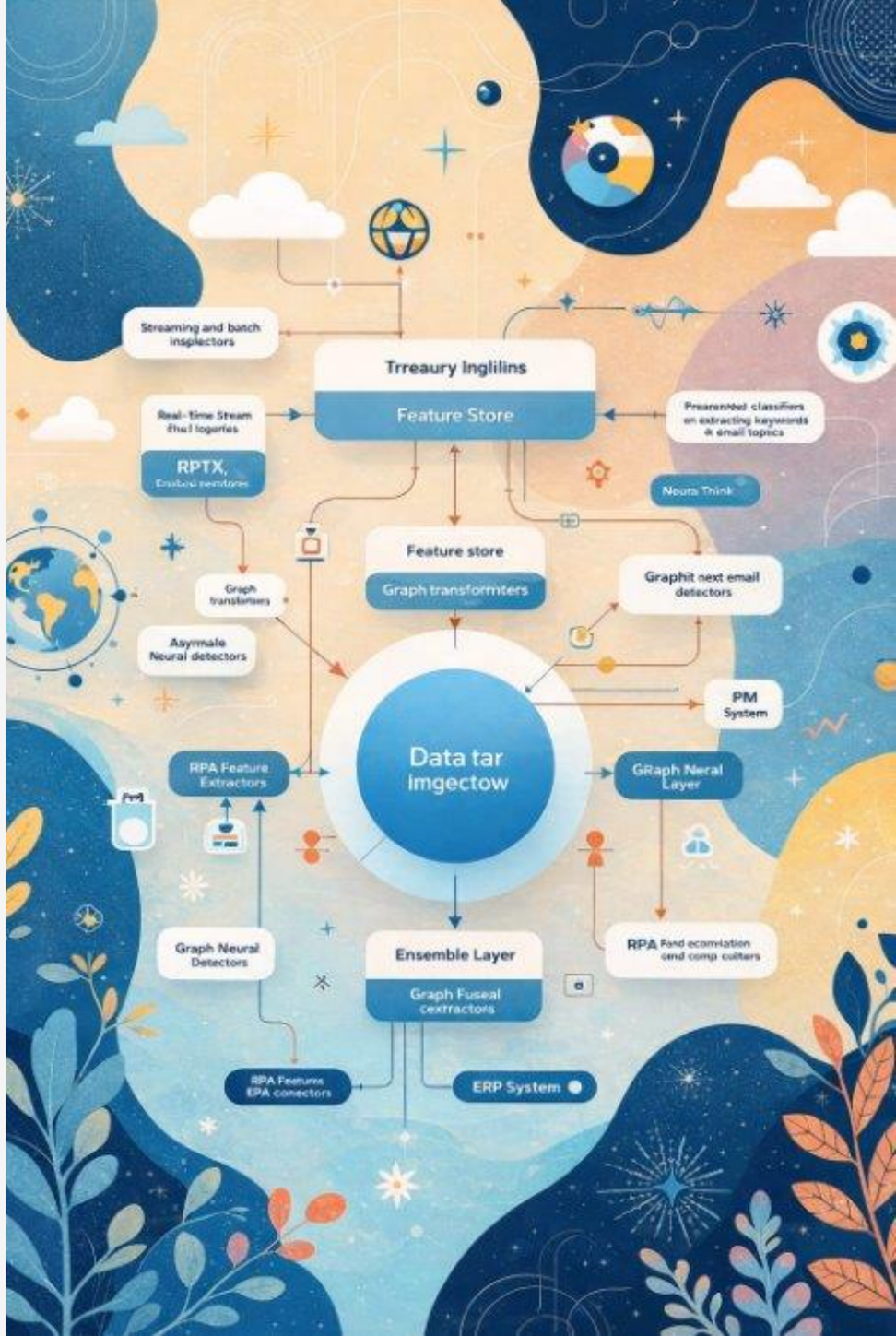
Ingestion & ETL: streaming and batch (Airflow/Kafka)

Feature store: temporal and graph features, entity resolution layer.

Models: supervised classifiers, unsupervised anomaly detectors, graph neural networks.

Fusion: ensemble scoring to combine transaction-level and network signals.

Automation: RPA for case creation; ERP connectors for audit trails (SAP/Oracle)



AI Architecture

- Data ingestion
- Streaming + batch processing
- Feature extraction
- Feature store
- Graph / network analytics
- Ensemble neural detectors
- Fusion layer
- ERP / RPA connectors
- Alerts / downstream systems

DataStreamer AI-First Data Pipeline Manager

Turn Chaos into Actionable Insights

Data Sources

200+ Sources

- Cloud Services
- SaaS Applications
- Network Devices
- Databases
- Endpoints
- Identity Systems

DataStreamer Core

1. AI-Powered Parsing
2. Threat Enrichment
   - Threat Intelligence Feeds
3. Detection at Edge
4. Auto-Scaling
5. Schema Evolution
6. Multi-Routing

AI

Attack
Malicious
Suspicious
Suspicious
MITRE ATTACK

MITRE
1M+ EPS

Real-Time Processing

Destinations

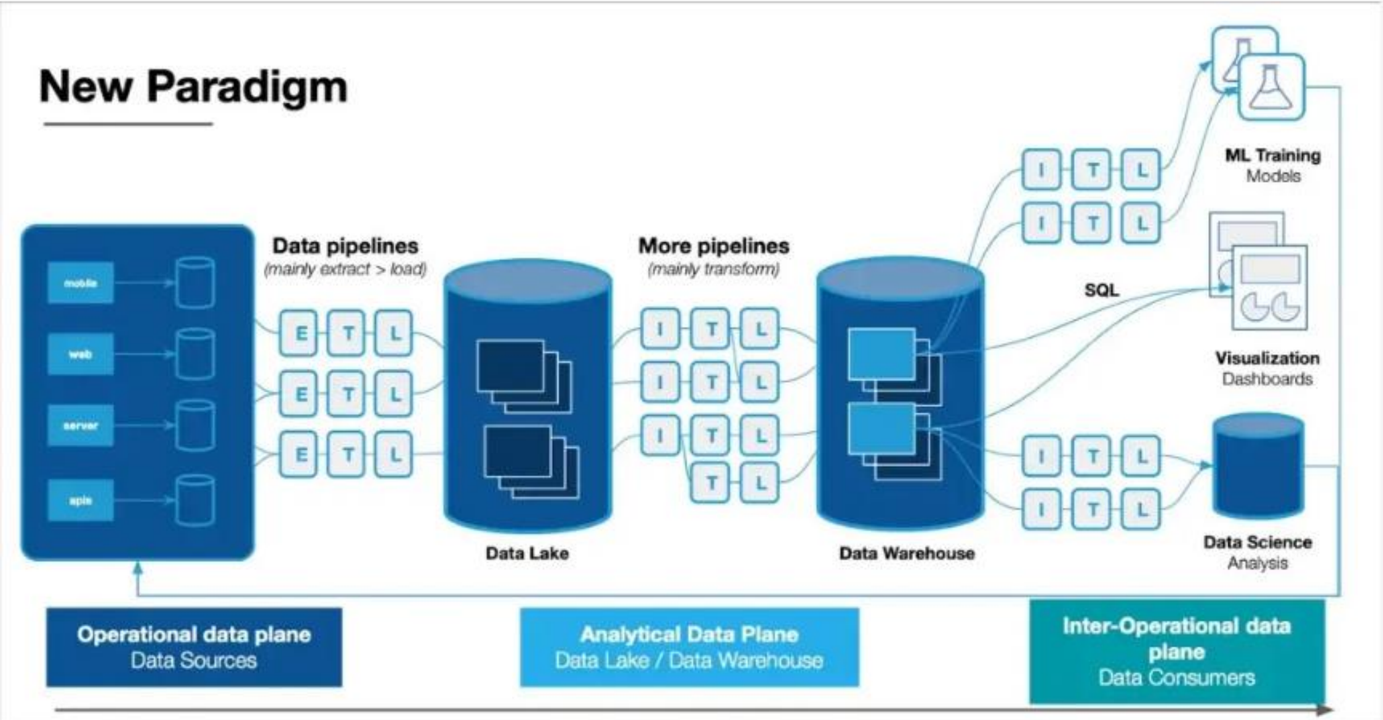
- SIEM Platforms
- Data Lakes
- Streaming Platforms
- Analytics Tools

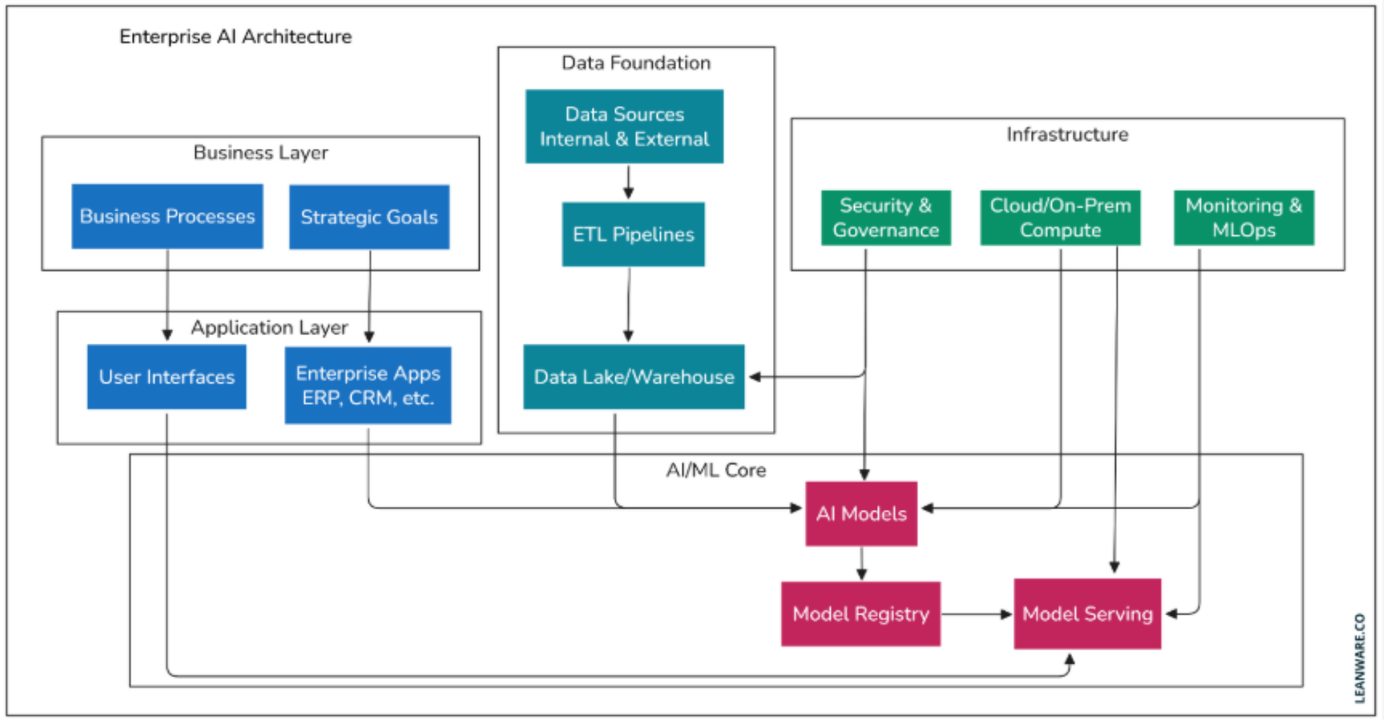
60% Cost Reduction

GDPR/SEBI/RBI Compliant

Zero Vendor Lock-in

AI-Ready Architecture


New Paradigm
mobile
web
server
apis
Data pipelines
(mainly extract > load)
E T L
E T L
E T L
Data Lake
More pipelines
(mainly transform)
I T L
I T L
I T L
T L
Data Warehouse
I T L
I T L
ML Training
Models
SQL
Visualization
Dashboards
I T L
I T L
Data Science
Analysis
Operational data plane
Data Sources
Analytical Data Plane
Data Lake / Data Warehouse
Inter-Operational data plane
Data Consumers

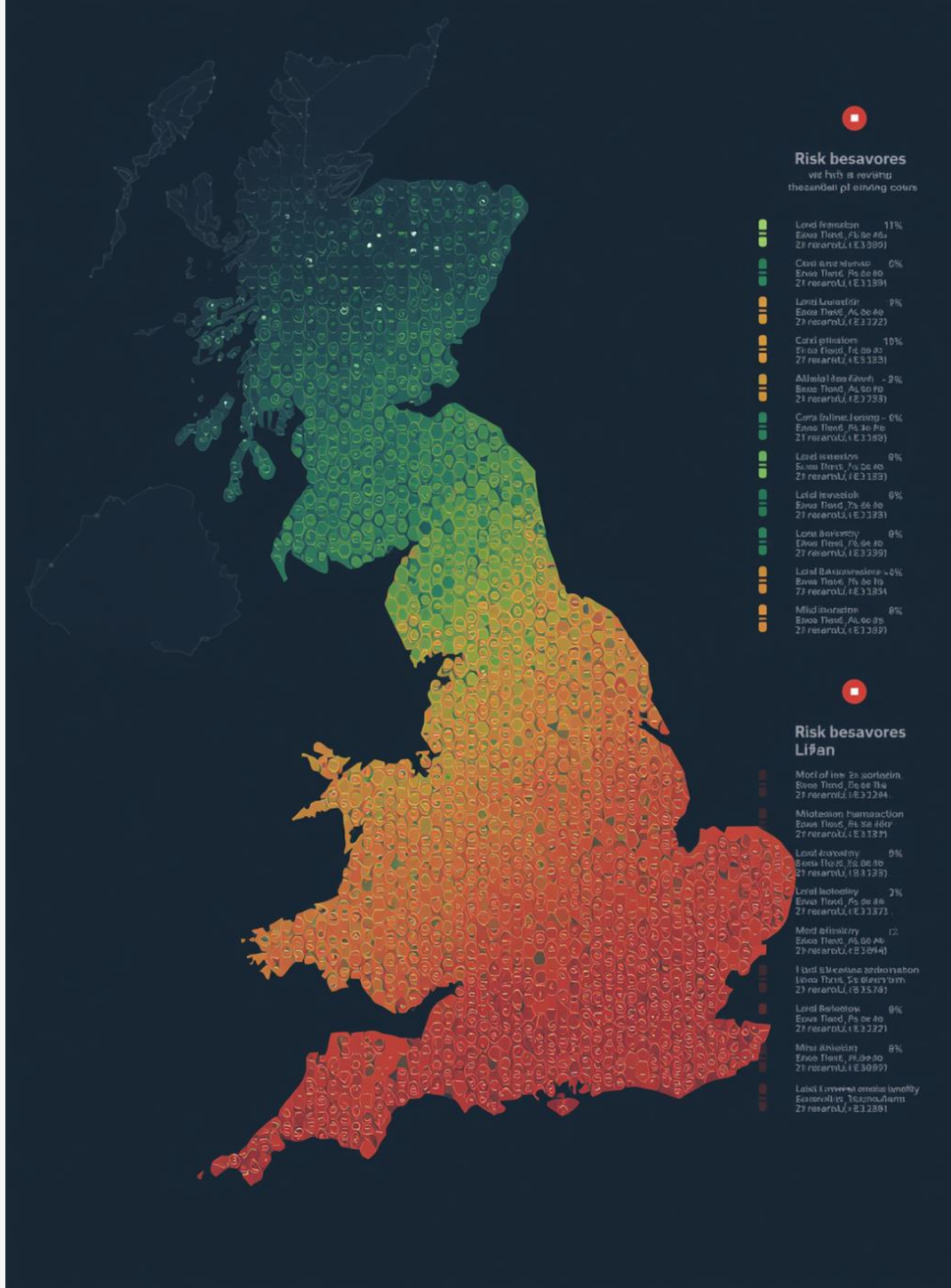

Enterprise AI Architecture
Data Foundation
Data Sources
Internal & External
ETL Pipelines
Data Lake/Warehouse
Business Layer
Business Processes
Strategic Goals
Infrastructure
Security &
Governance
Cloud/On-Prem
Compute
Monitoring &
MLOps
Application Layer
User Interfaces
Enterprise Apps
ERP, CRM, etc.
AI/ML Core
AI Models
Model Registry
Model Serving
LEANWARE.CO

Analytical Techniques & Results

The hybrid model delivered: 91% accuracy vs 65% for rules; 40% fewer false positives compared to legacy systems. Graph analytics uncovered multi-account laundering rings missed by transaction-only models.

Risk Scoring Model

The entity-level risk scoring model evaluates transaction behavior and anomalies to assign risk scores, enhancing the detection of potential money laundering activities effectively.

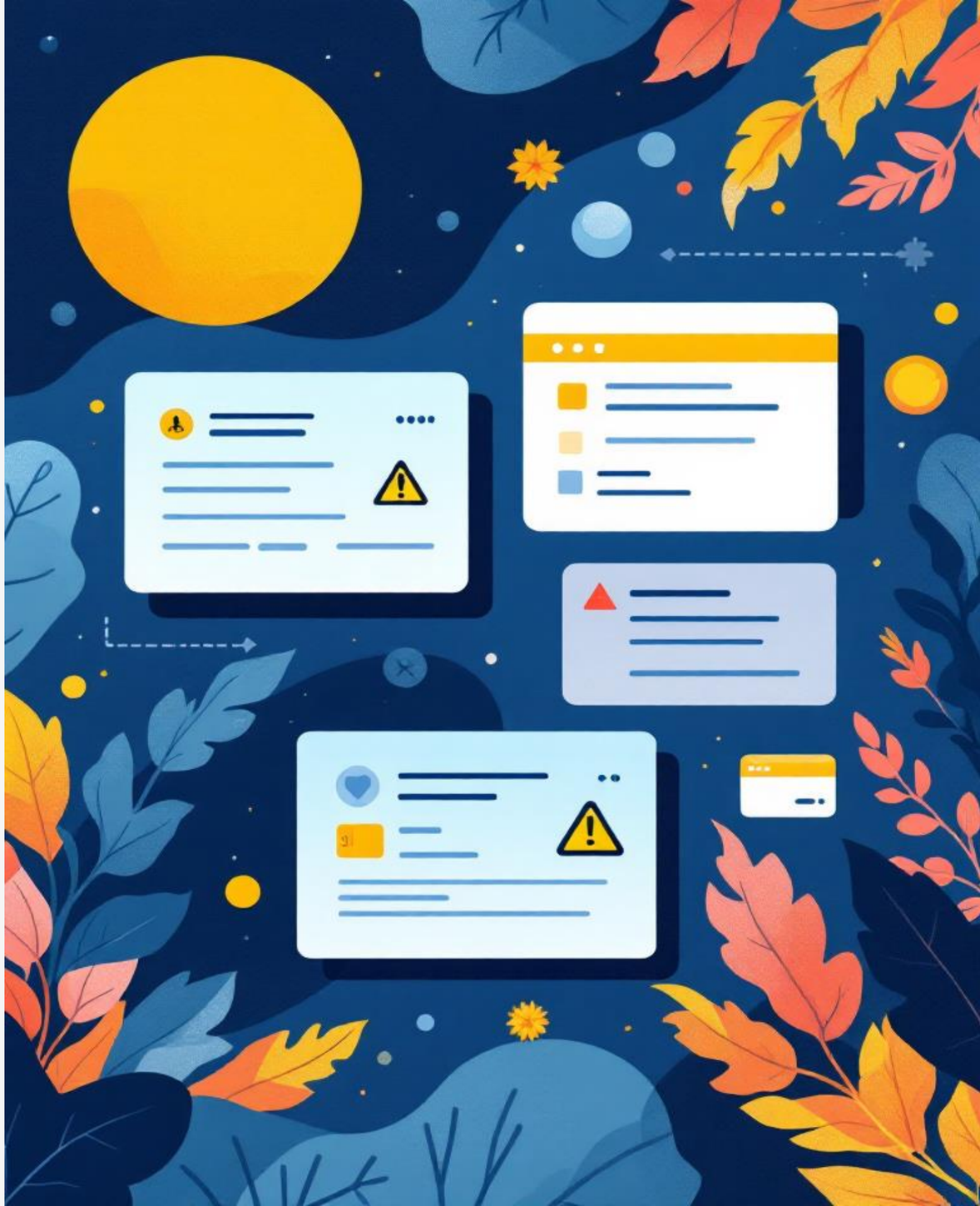


Key Performance Indicators

This section outlines the **essential metrics** for evaluating the performance of Anti-Money Laundering systems, focusing on detection rates, false positives, and overall operational efficiency.

Fraud Detection Rates

This section illustrates the **varying detection rates** of fraud across different transaction types, highlighting which categories exhibit the highest levels of risk and the need for tailored strategies.

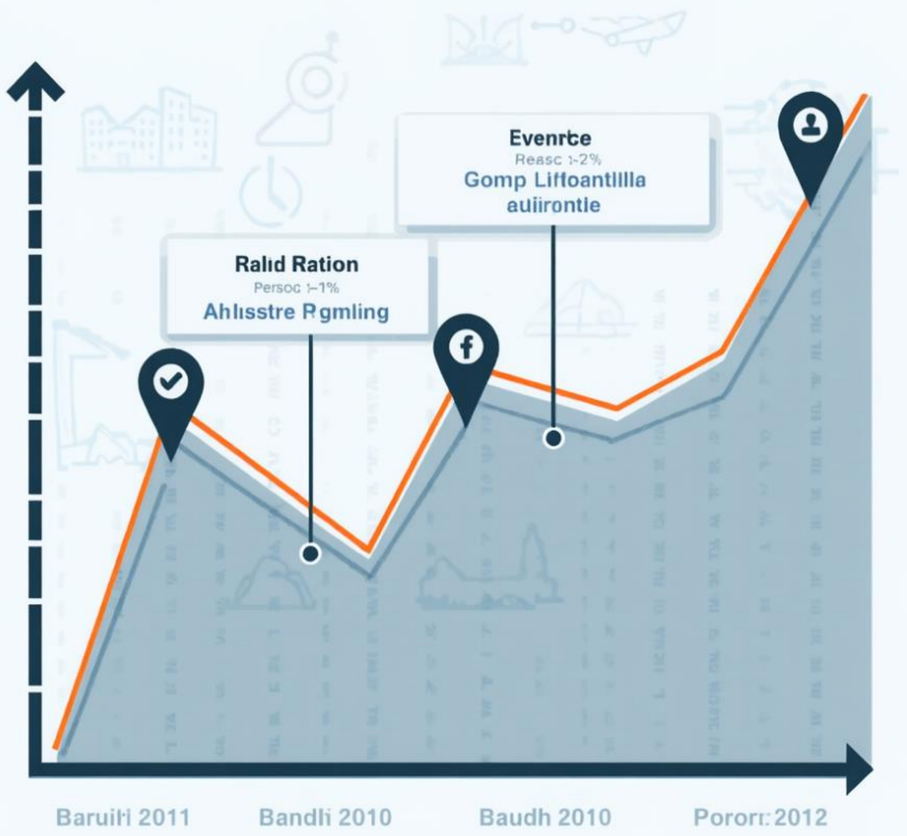


Entity Risk Scoring

This section analyzes **transaction amounts** and behaviors to determine risk levels. The scatter plot illustrates patterns that help identify high-risk entities based on their transaction activities.

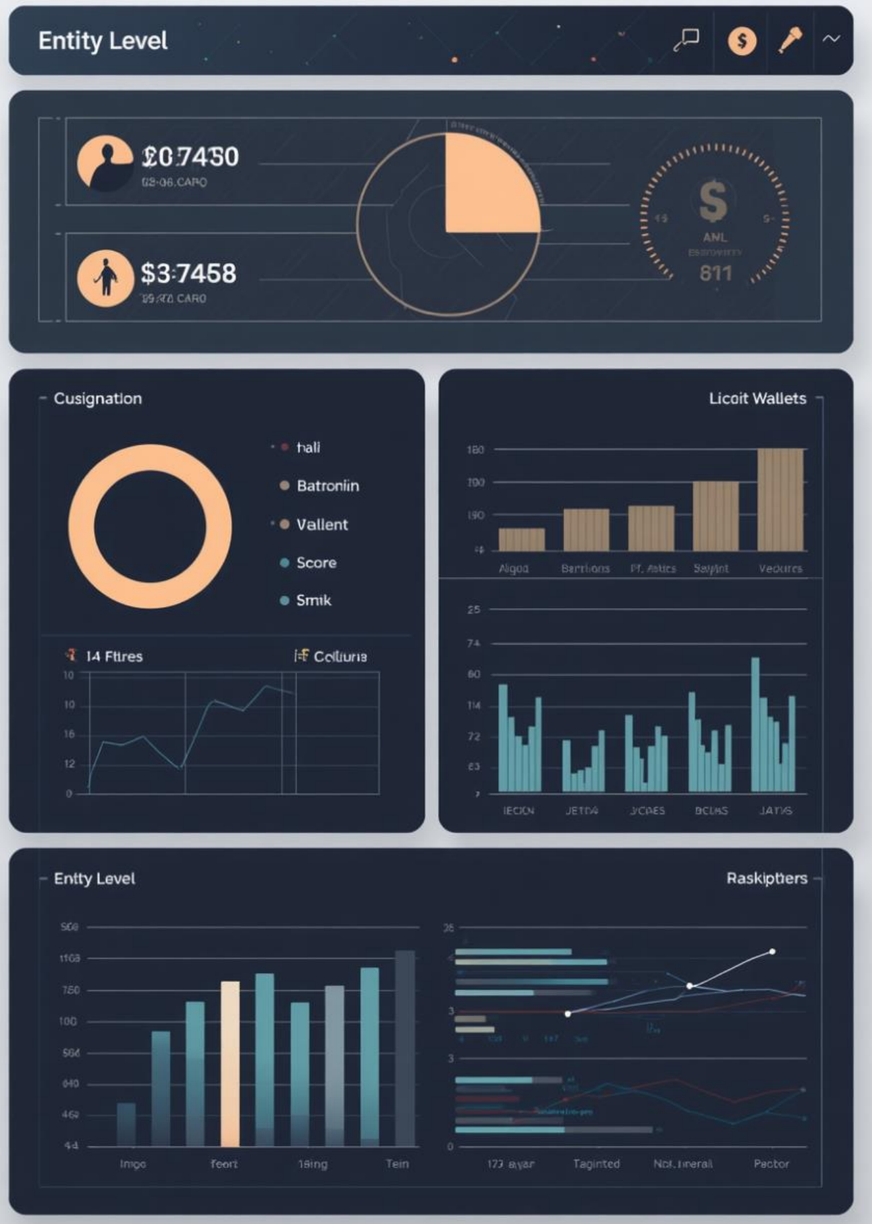
Temporal Fraud Trends

This section highlights the **dynamics of fraudulent transactions** over time, illustrating how fraud detection evolves. The time-series graph captures key events and trends in fraudulent activity.



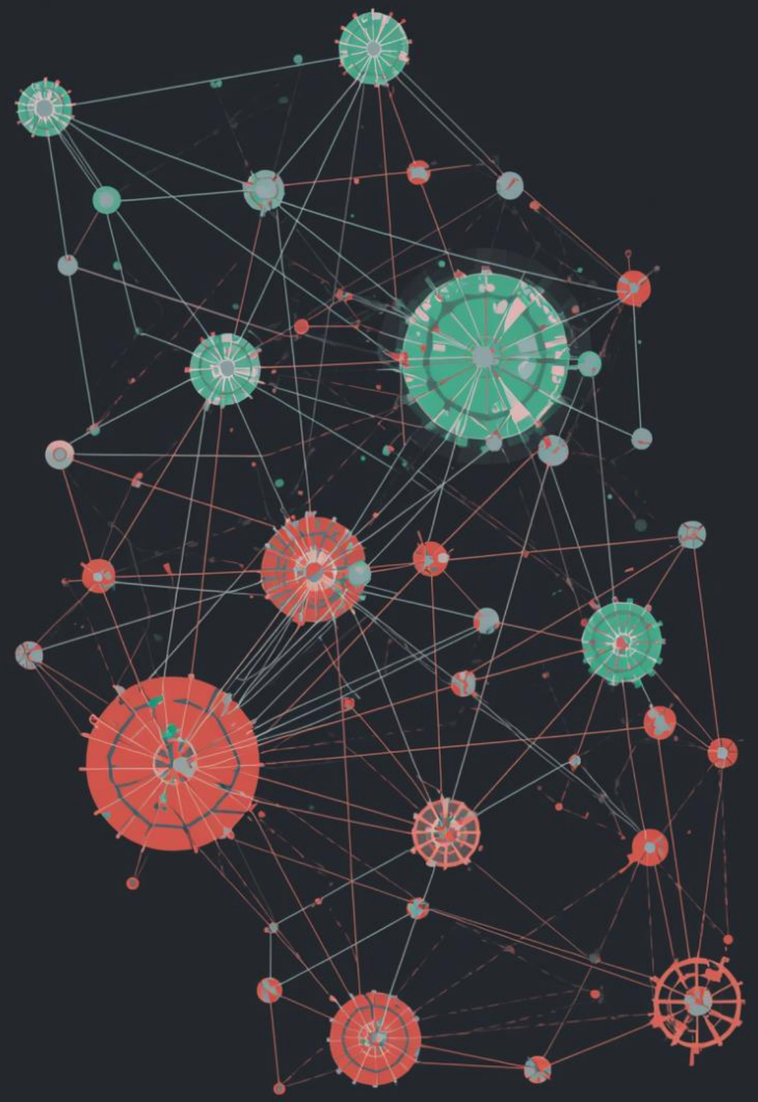
Entity-Level Dashboard

This section presents the **entity-level AML dashboard**, show casing the distribution of illicit wallets and overall performance metrics, enhancing our understanding of compliance and detection effectiveness.



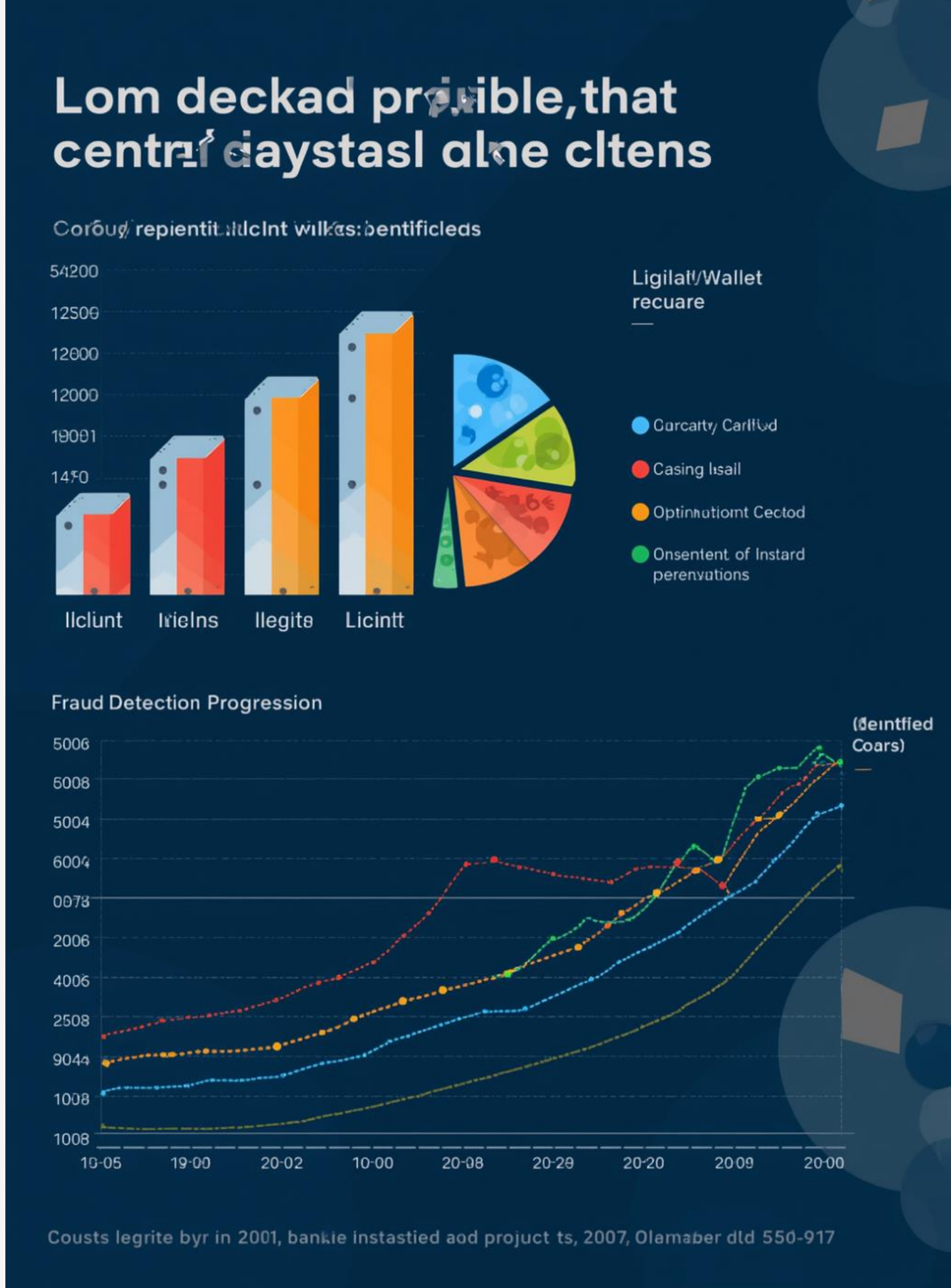
Clustering & Network Analysis

This section explores how **feature-based clustering** separates licit and illicit wallets, enhancing detection accuracy by revealing hidden connections within the financial networks.



Illicit Wallets Overview

This section highlights the identification of illicit wallets and trends in fraud detection, show casing the effectiveness of AI-driven methodologies in combating money laundering over time.



Entity Resolution & Graph Use Cases

Circular Flows

Detect circular transfers and layering by aggregating weak signals across accounts.

Hub-and-Spoke Rings

Identify central facilitators: high-degree nodes that route funds between many accounts.

Entity-Level Scoring

Combine behavioral anomalies, centrality, and proximity to illicit labels into a single risk score.

ERP Integration & Operationalization

Seamless ERP integration (SAP S/4HANA, Oracle Financials) enables: - Real-time transaction enrichment and automated posting of alerts. - Audit trails and immutable logs for regulatory review. - Closed-loop workflows: model output → RPA case creation → investigator UI → feedback to models. Recommended: ISO 27001 controls, data lineage, and role-based access.

Risks, Ethics & Limitations

Data Bias & Explainability
Entity scores risk embedding historical bias; require XAI, model cards, and human-in-the-loop review.

Privacy & Compliance
GDPR and data minimization constraints demand careful feature selection and pseudonymization.

Operational Fragility
Pipeline failures (ingest/orchestration) propagate errors — implement monitoring, retries, and canaries.

Compute & Data Quality
GNNs are resource-intensive and sensitive to poor entity resolution; invest in data engineering and indexing.

Recommendations & Next Steps

1. Pilot & Validate

Run a scoped pilot combining PaySim+Elliptic patterns with production data; measure precision, recall, and investigator time saved.

2. Governance Adopt responsible AI policies:

bias audits, explainability, logging, and SRO oversight.

3. Scale & Integrate

Operationalize via robust ETL, feature store, ERP connectors, and RPA for case management.

4. Future Proof

Explore federated learning, blockchain-based provenance, and continuous retraining to counter adversarial adaptation.

Thank You